

Exploratory Data Analysis: Bronx Taxi Travel Patterns

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April 13 2025

1 Data Sources, Preparation, and Cleaning

Through literature review, we wanted to identify taxi hubs for NYC. 2018 NYC Taxi data was pulled to measure length of trips, pickup and drop-off locations, average fare cost, and many more. The purpose of these data was to take a broad look at the bigger picture. This dataset included more than one million entries. [8]

Preparation of these data involved R, including running summary statistics, comparison bar graphs and line graphs, as well as mutating the data in order to take more than 270 individual location codes regarding pickup and drop-off locations, into the proper regions that the codes belonged to.

Our analysis utilizes datasets released by the NYC Taxi and Limousine Commission (TLC), including High Volume For-Hire Vehicle (HVFHV), Yellow Taxi, Green Taxi, and traditional For-Hire Vehicle (FHV) trip data for January 2025. Additionally, we use a GeoJSON file from the TLC defining New York City's taxi zones, which includes zone boundaries, names, and boroughs.[8]

Another part of our analysis utilizes the 2023 Disadvantaged Communities (DAC) Census Tract Dataset, published by the New York State Climate Action Council. The dataset flags census tracts across New York as either "Designated as DAC" or "Not Designated" based on multiple indicators of environmental burden and social vulnerability. In addition to this designation, the dataset includes over 60 socioeconomic and environmental features, such as population demographics, internet access, educational attainment, and exposure to environmental hazards.[2]

To focus the analysis on New York City, we joined this dataset with 2020 TIGER/Line Shapefiles from the U.S. Census Bureau, filtering by FIPS codes to extract census tracts within the five NYC counties (Bronx, Brooklyn, Manhattan, Queens, and Staten Island). Both files were cleaned and standardized, ensuring the GEOID column was consistent for merging. [8][9]

During preprocessing, we removed corrupted rows where location values (e.g., County, City_Town) were incorrectly replaced by latitude and longitude coordinates. These entries could not be recovered and were dropped. The `Urban_Rural` column had a few missing values, which were imputed as "Urban" since all five boroughs fall within urbanized areas. Numeric columns containing missing values were filled with 0 to ensure compatibility for aggregation, visualization, and modeling.

2 Early Analysis

Early in the analysis, and since there were so many entries, we needed to breakdown the data into smaller parts to look for relationships. One of our early EDA involved looking into the average length of each taxi ride. It was evident that trips averaged roughly between six and twelve miles from the graph below.

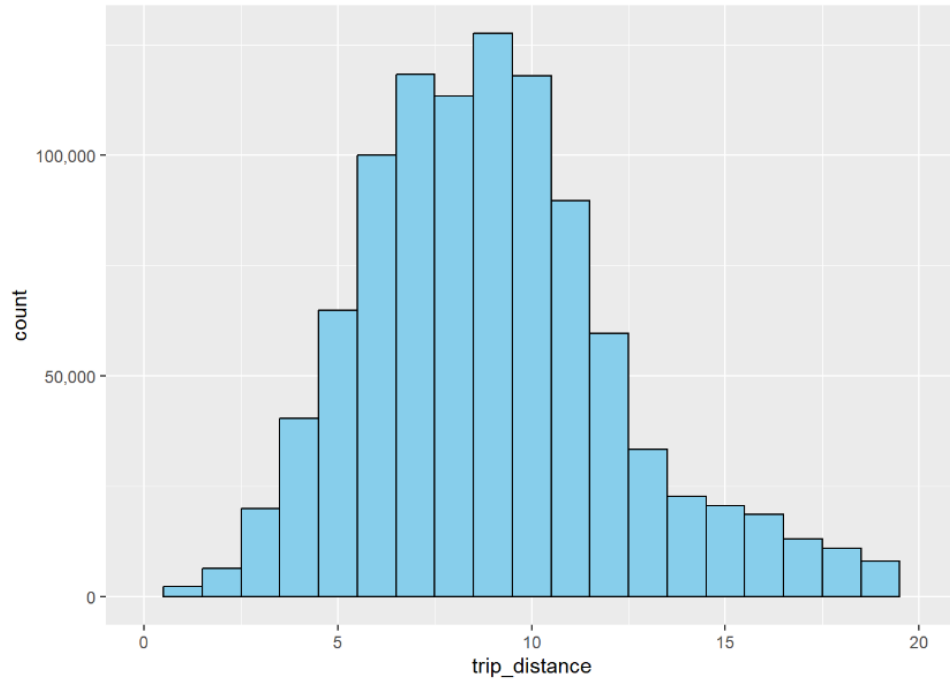


Figure 1: Histogram of Average Trip Distance.

After this we wanted to look into pickup and drop-off locations. To do this, we compared each location type to average trip distance to see if there was any conclusion to be made. It was clear that there were some outliers, even though only having the location codes did not give us the particular regions.

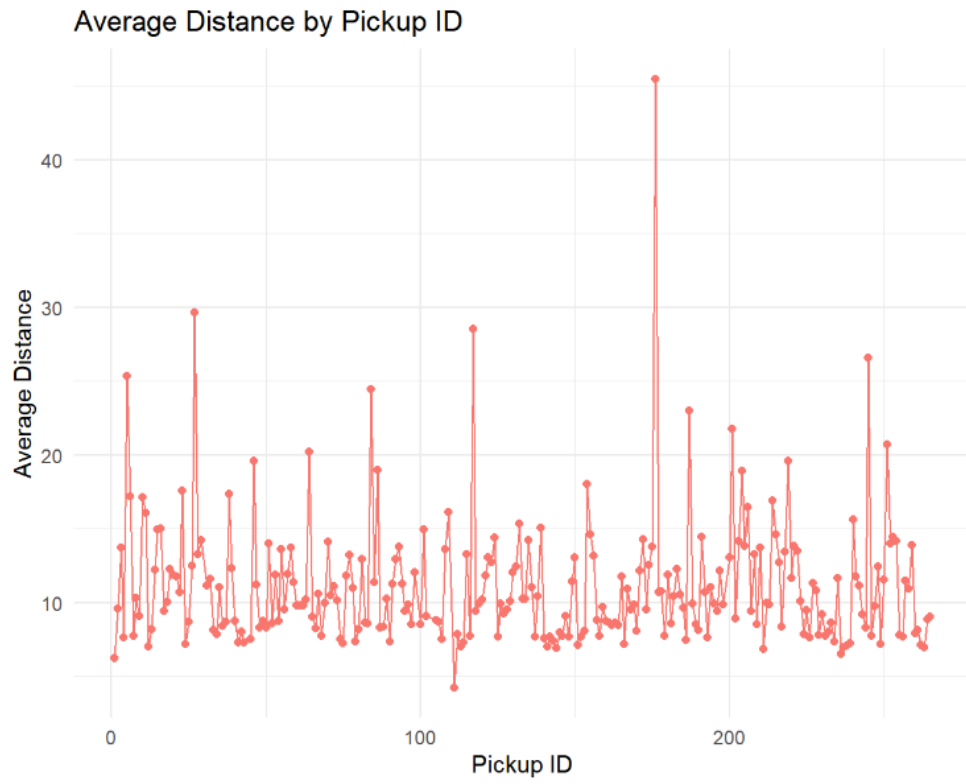


Figure 2: Average Trip Distance by Pickup Code.

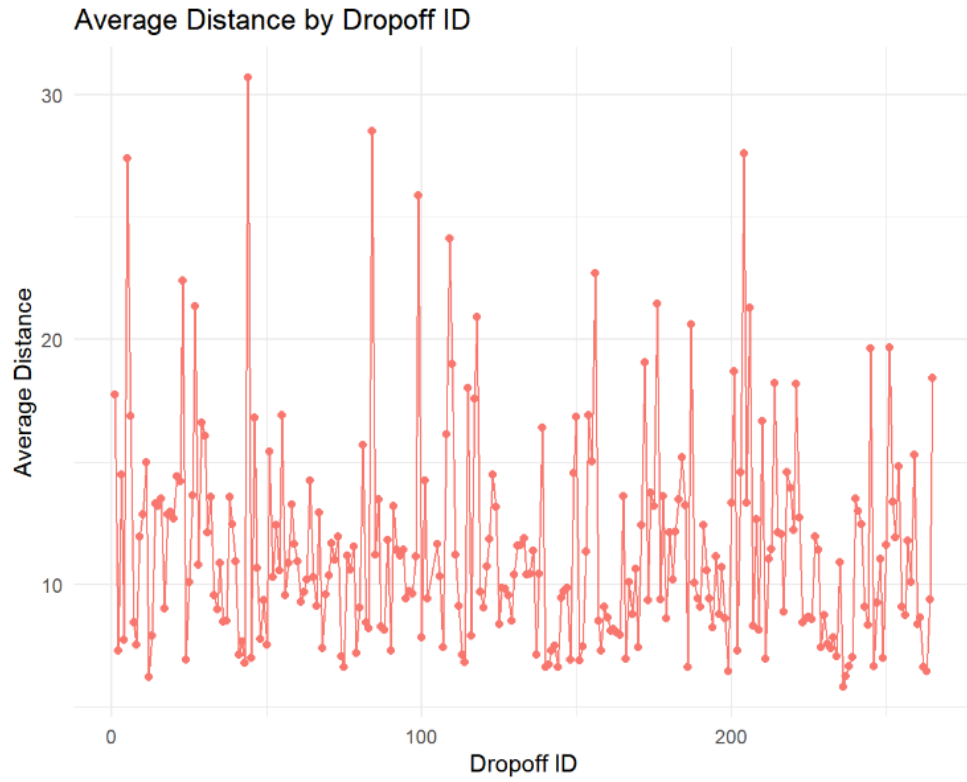


Figure 3: Average Trip Distance by Drop-off Code.

The next objective was to create visualizations of the number of trips for each location code. This was done by comparing the number of trips to each code and placed into bar graphs for pickup and drop-off codes, respectively.

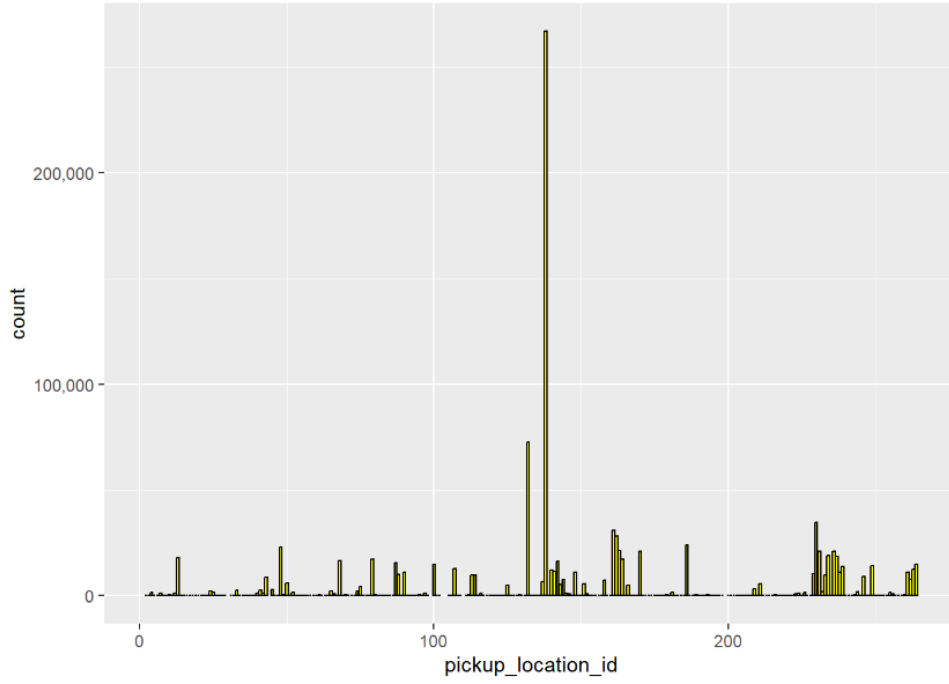


Figure 4: Number of Pickups for each Location Code.

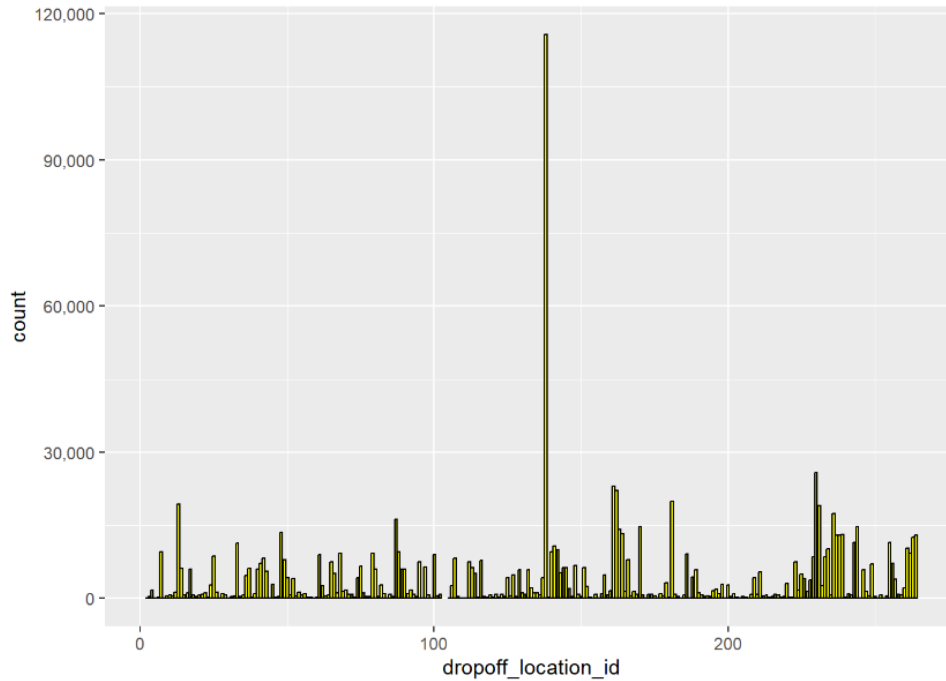


Figure 5: Number of Drop-offs for each Location Code.

However, just having the location codes was not going to give us the answers we were looking for. We needed to take these codes and give them names to represent the regions of NYC: Bronx, Brooklyn, Manhattan, Queen, and Staten Island. To do this, the R code was mutated to place each code into a vector

so that it could be labeled with its appropriate borough. Then, a bar graph was created to display the number of trips for each borough, which gave us a clear understanding of where taxis start and finish the vast majority of their trips.

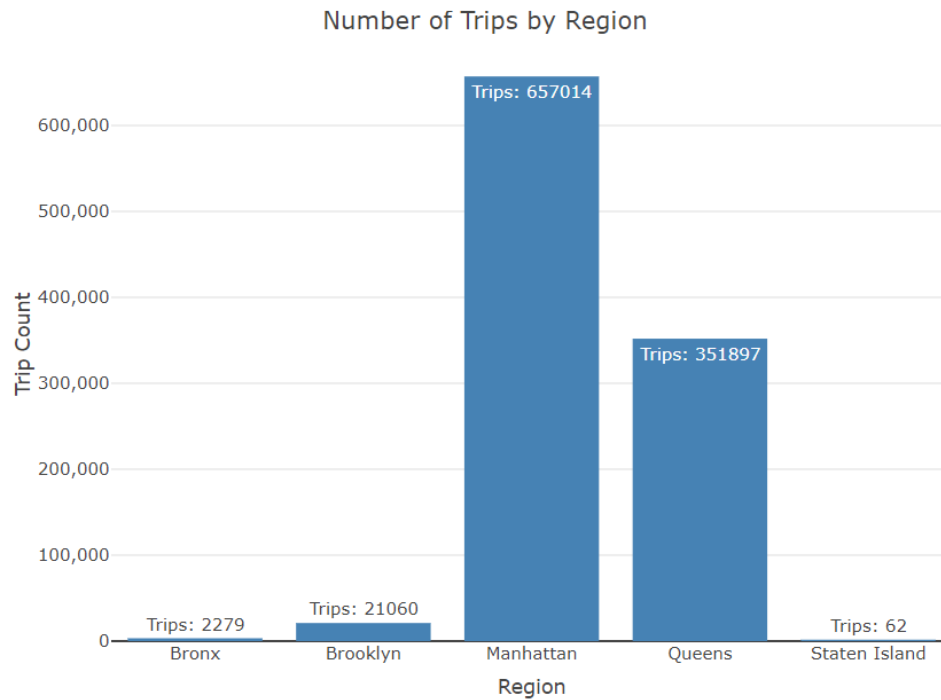


Figure 6: Bar Graph Representing Taxi Trips in each Borough.

Since the data also included hour of the day, day of the week, and total cost of each trip we wanted to then see if there was a relationship with those variables by running exploratory summary statistics. Here, we can see each region's most common day for taxi rides, average cost, and the most common hour of the day that rides were taken.

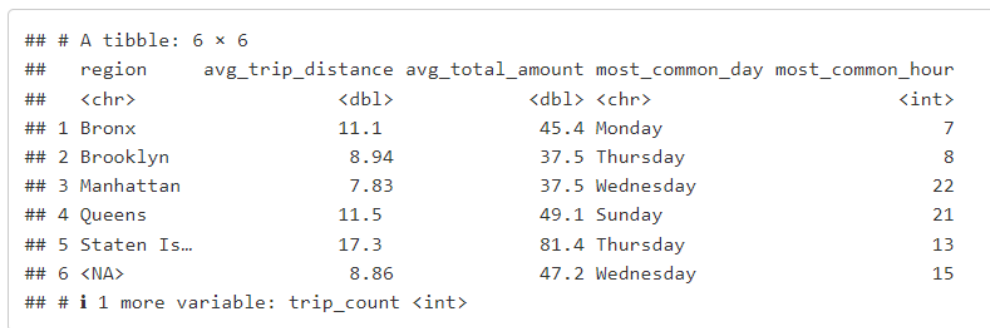


Figure 7: Summary Statistics.

We then wanted to look at the relationship between day of the week and hour of the day. To do this, we created a heatmap comparing these two variables.

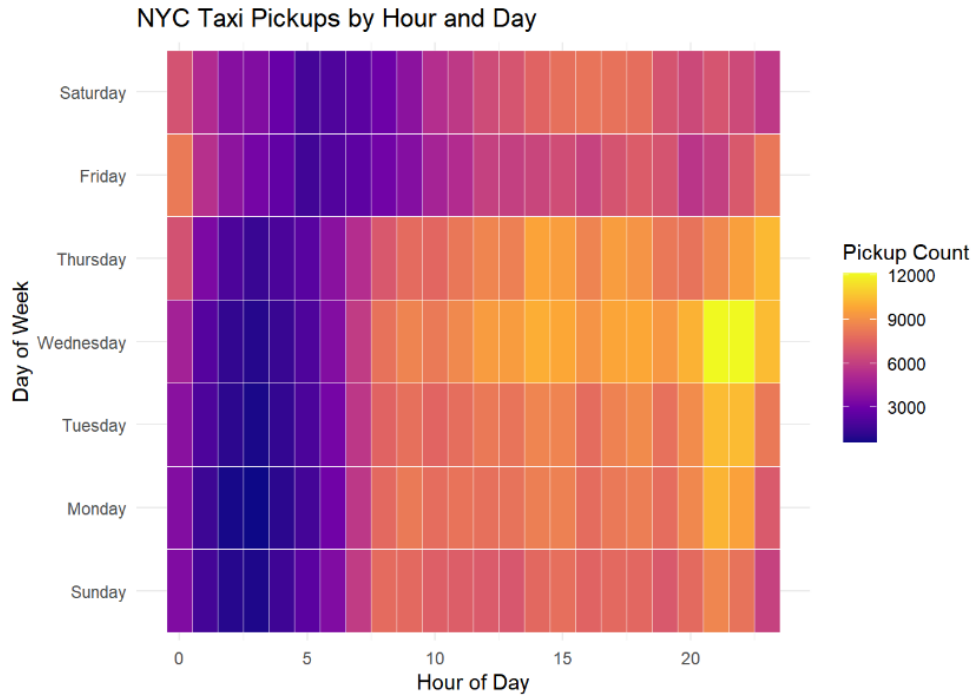


Figure 8: Heatmap of Hour of the Day and Day of the Week.

3 Evaluating our target locations and demographics

However, we still need to make decisions on what groups in NYC are lacking in public transportation. So-called "transit deserts".

3.1 DAC Tract Distribution by Borough

We began our exploratory analysis by examining the geographic distribution of DAC census tracts across the five boroughs of New York City. A stacked bar chart revealed that the Bronx has the highest absolute number of DAC-designated tracts, followed closely by Brooklyn and Queens. While Manhattan and Staten Island have significantly fewer DAC tracts, the relative proportion remains noteworthy, particularly given differences in land area and population density.

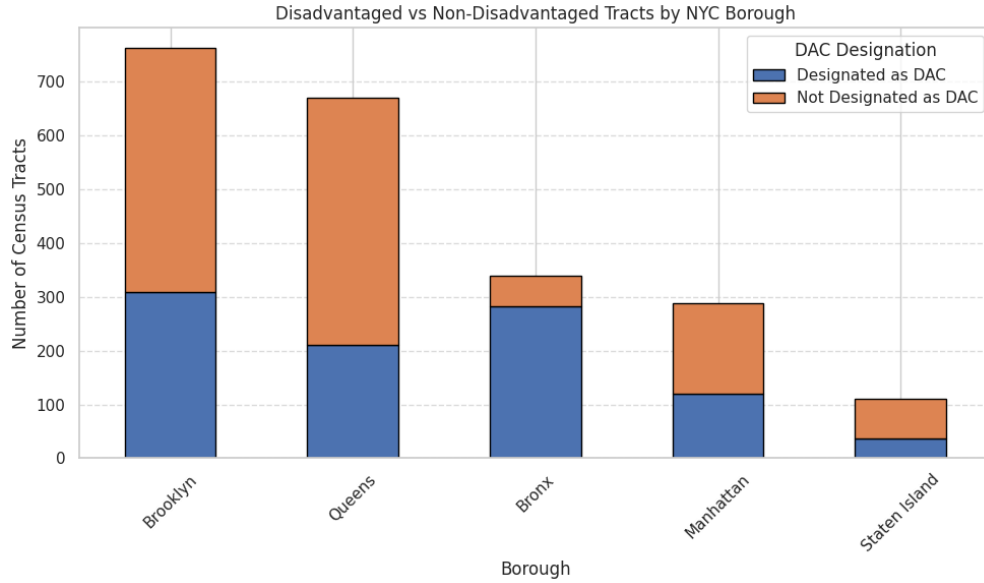


Figure 9: Disadvantaged vs Non-Disadvantaged Tracts by NYC Borough

To further investigate these burdens, we used boxplots and kernel density plots to compare socioeconomic indicators between DAC and Non-DAC tracts. Interestingly, Internet Access was higher on average in DAC tracts, with a tighter distribution centered around 70–90 percent. In contrast, Non-DAC tracts exhibited a lower median and wider spread, with many areas reporting very low access. While counter to initial expectations, this may reflect concentrated investments in broadband infrastructure in underserved areas, or outliers in higher-income zones where alternative living arrangements reduce connectivity metrics.

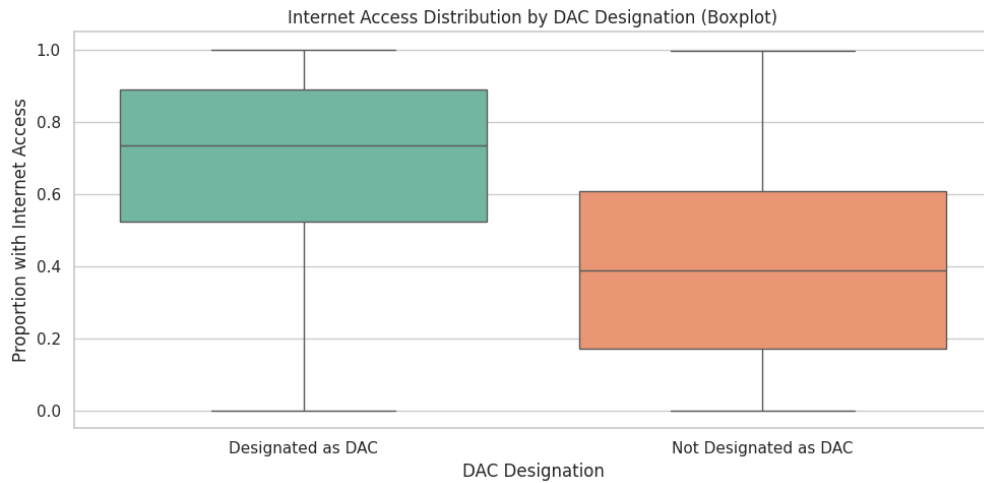


Figure 10: Internet Access Distribution by DAC Designation

Additionally, we visualized three housing-related attributes—Rent as a Percentage of Income, Proportion of Renters, and Home Energy Affordability. Across all three variables, DAC tracts consistently showed higher median burdens and broader distributions. Renters in DAC communities tend to allocate a larger share of income to housing, and more frequently experience energy-related financial strain. These patterns are visually evident in both the boxplots and the KDE curves, which shift rightward for DAC tracts in all cases.

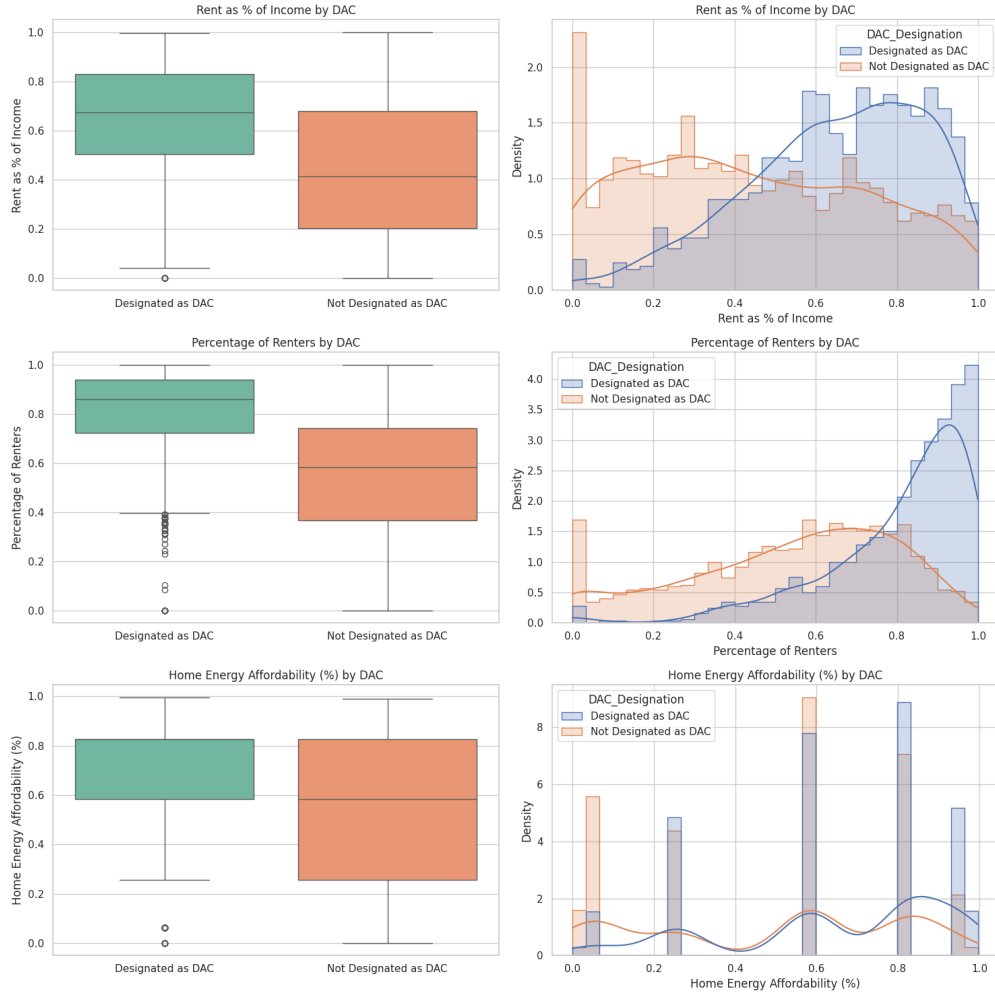


Figure 11: Rent Burden Tri-Graph by DAC Designation

3.2 Educational Attainment

The choropleth map visualizing Population with No College Degree revealed strong spatial alignment with DAC-designated tracts. Higher proportions of residents without college degrees were especially concentrated in areas such as Central Brooklyn, Northern Manhattan, and the South Bronx. The visual gradient of the map made it evident that educational attainment disparities are geographically clustered and disproportionately impact DAC communities across New York City.

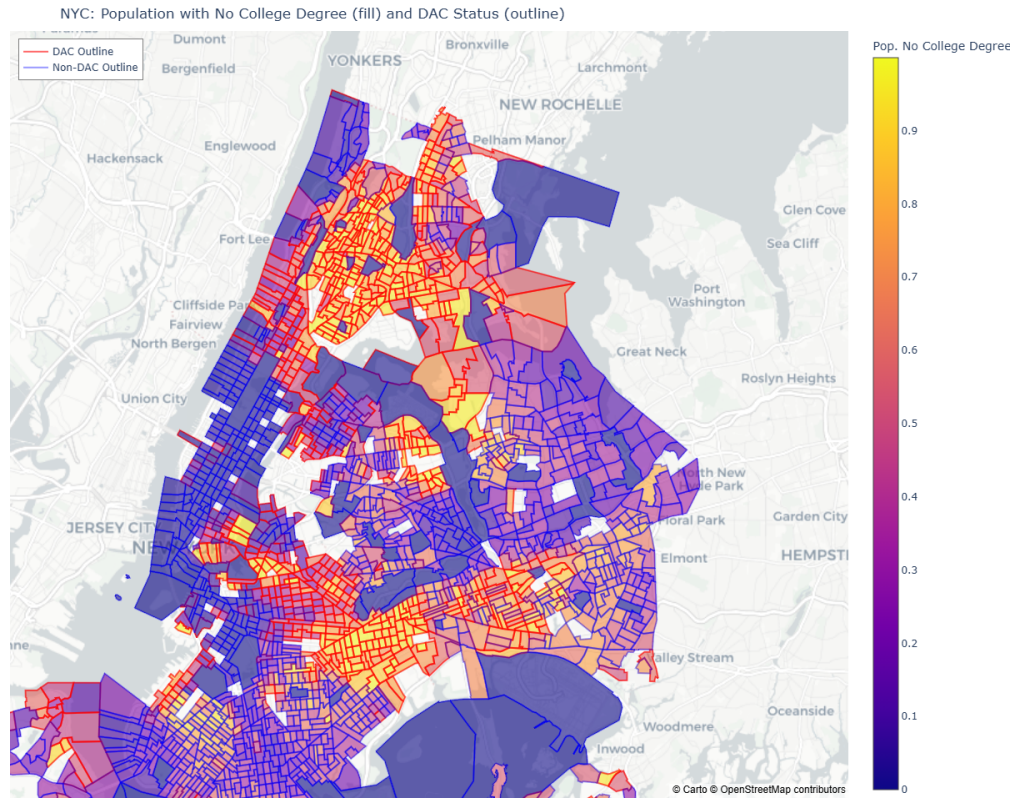


Figure 12: NYC: Population with No College Degree (Fill) and DAC Status (Outline)

3.3 Vulnerability and Health Outcomes

Choropleths of Unemployment Rate and Households with Disabilities showed strong spatial overlap with DAC areas, indicating that designated tracts often contain more disabled residents and experience higher unemployment. In contrast, the Residents Over Age 65 choropleth revealed a more nuanced pattern—many DAC tracts have lower proportions of older residents than Non-DAC areas. This suggests that vulnerability in DAC-designated tracts is not necessarily linked to age, but more likely tied to economic hardship, disability prevalence, and other systemic disadvantages.

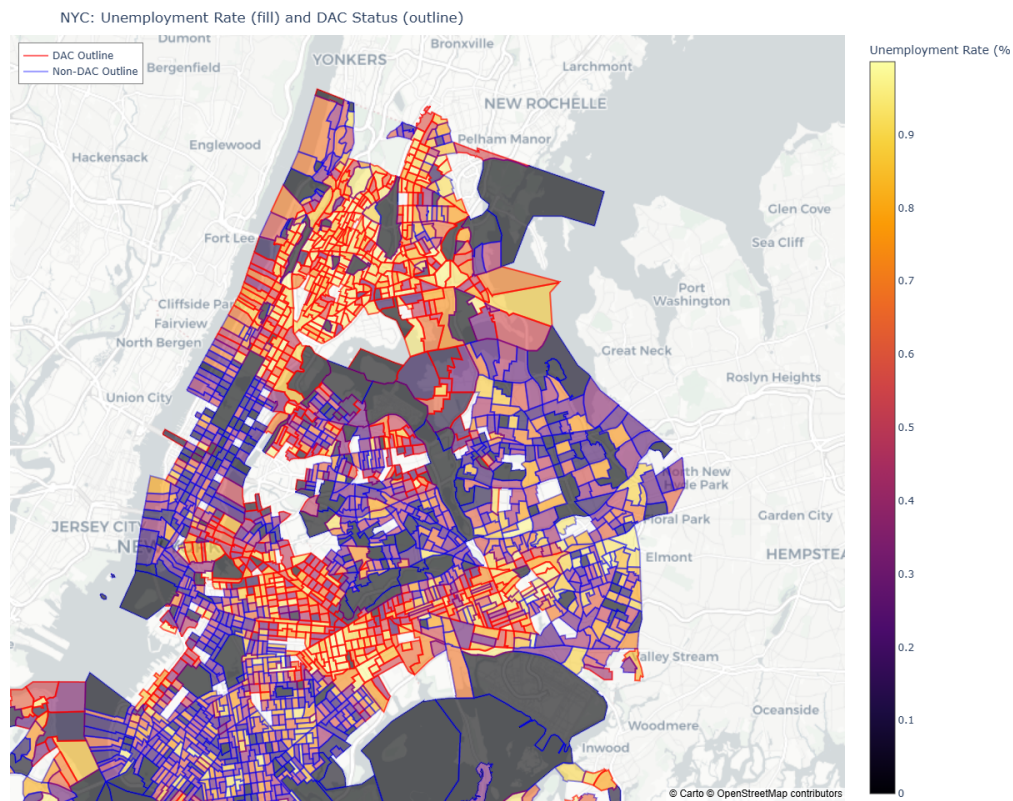


Figure 13: NYC: Unemployment Rate (Fill) and DAC Status (Outline)

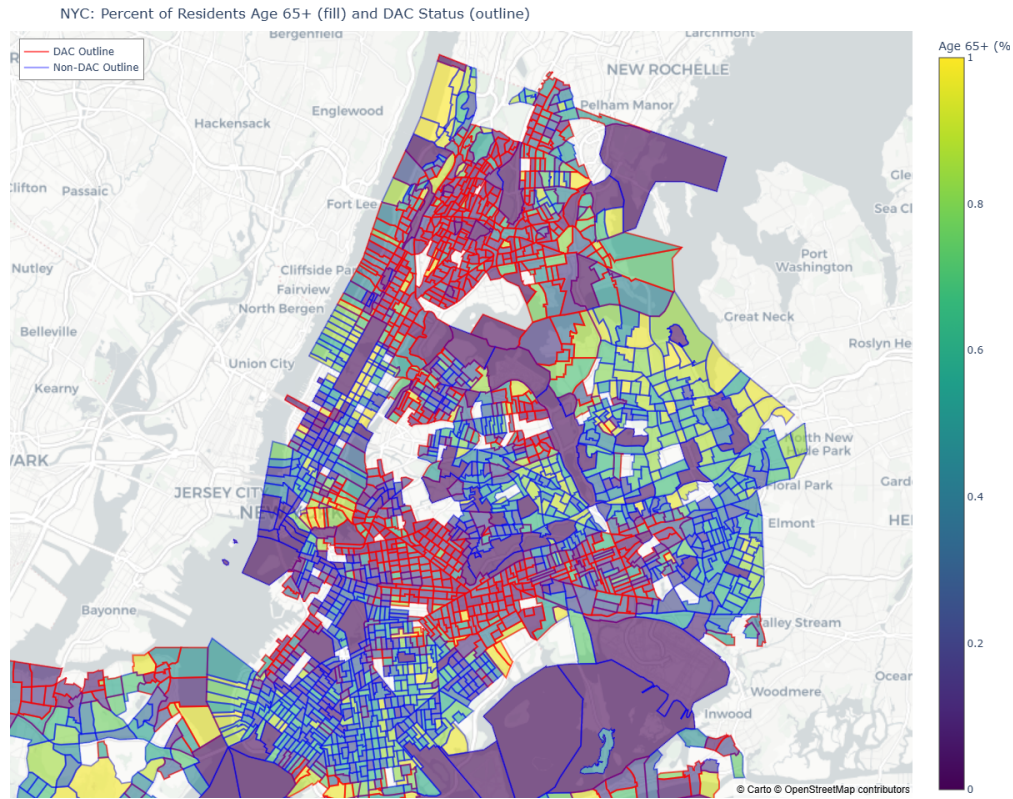


Figure 15: NYC: Percent of Residents Age 65+ (Fill) and DAC Status (Outline)

3.4 Multivariable Patterns

Using a full correlation heatmap across cleaned numeric columns, we observed strong positive relationships between indicators like No College Degree, Unemployment, and Disability, while Internet Access negatively correlated with many vulnerability and burden metrics. This visualization emphasized how social, economic, and environmental disadvantages frequently cluster within DAC communities.

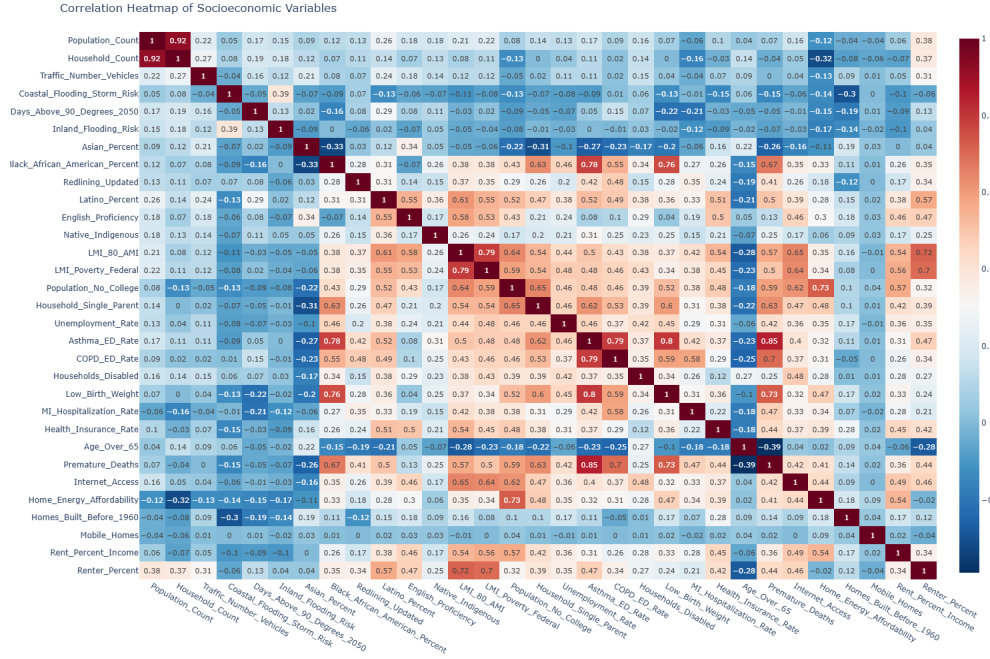


Figure 16: Correlation Heatmap of Socioeconomic Variables

Based on our overall analysis of the DAC, we discovered the majority of transit deserts are located in the Bronx, to simplify the analysis, and combine with other datasets more easily. We use "The Bronx" as our analysis, rather than specific PU location IDs.

4 Trip Filtering and Feature Engineering

To examine travel behavior specific to the Bronx, we filtered all trips that either started *or* ended in the Bronx. This gave us a borough-centric lens to study travel dynamics and infrastructure dependencies. Coordinates for each pickup and dropoff zone were extracted from the centroid of each polygon in the shapefile, projected to latitude and longitude using EPSG:4326.

After filtering, trips were aggregated by origin, destination, and vendor type. This allowed for both vendor-based visualization and analysis of spatial patterns in mobility.

For the 2023 Disadvantaged Communities (DAC) Census Tract Dataset, following initial cleaning, we engineered a variety of enhancements to prepare the data for analysis. Numeric fields were converted to float type and checked for consistency using summary statistics and visual inspection, and subsets for DAC vs. Non-DAC tracts were created to support comparative analysis.

Spatially, the merged dataset was transformed into a GeoDataFrame using tract-level geometries. This enabled the creation of choropleth maps with outline overlays for DAC and Non-DAC areas using Plotly's Scattermapbox and Choroplethmapbox layers. We also developed a custom Borough column, which proved useful for aggregations and for interpreting borough-specific trends.

5 Key Findings

5.1 Methods

To perform borough-level aggregations, the GeoJSON's `LocationID` field was used to join with the pickup and dropoff location IDs in the Parquet trip files. The data was cleaned to remove nulls and formatted consistently across all sources. Each file was individually standardized to include:

- A unified `Pickup Datetime` and `Dropoff Datetime`

- PULocationID, DOLocationID, and corresponding PU Borough, DO Borough
- A new column named **Vendor** mapping provider codes to labels (e.g., “HV0003” → “Uber”)
- A **fare** column aggregating all passenger-paid costs (fare, tips, taxes, surcharges, etc.)

Each dataset was processed using Python (Pandas and GeoPandas), and concatenated into a unified file `AllTrips.parquet`, containing over 2 million records. To visualize spatial trends in Bronx taxi data, we utilized **Kepler.gl**, an open-source geospatial analysis tool developed by Uber. Kepler.gl allows users to create interactive maps from large-scale location datasets using simple drag-and-drop CSV uploads. After cleaning and aggregating our data in Python, we exported a CSV containing origin and destination coordinates, trip counts, and vendor labels. This file was then imported into Kepler.gl, where we configured *Arc Layers* to represent the flow of taxi trips between boroughs. By customizing stroke width and color based on trip volume and vendor, Kepler enabled us to clearly highlight key spatial insights—most notably, the dominance of long-distance trips from the Bronx to NYC airports and the clustering of local trips around major Hudson River crossings. Kepler’s interactivity and visual clarity made it a powerful tool for communicating these high-level geographic patterns in our final exploratory analysis.

5.2 Airports as Long-Distance Anchors

One of the most striking observations is that the **most frequent long-distance trips from the Bronx terminate at the region’s airports**—particularly LaGuardia and JFK. These trips represent the longest average distances and highest fares originating in the Bronx.

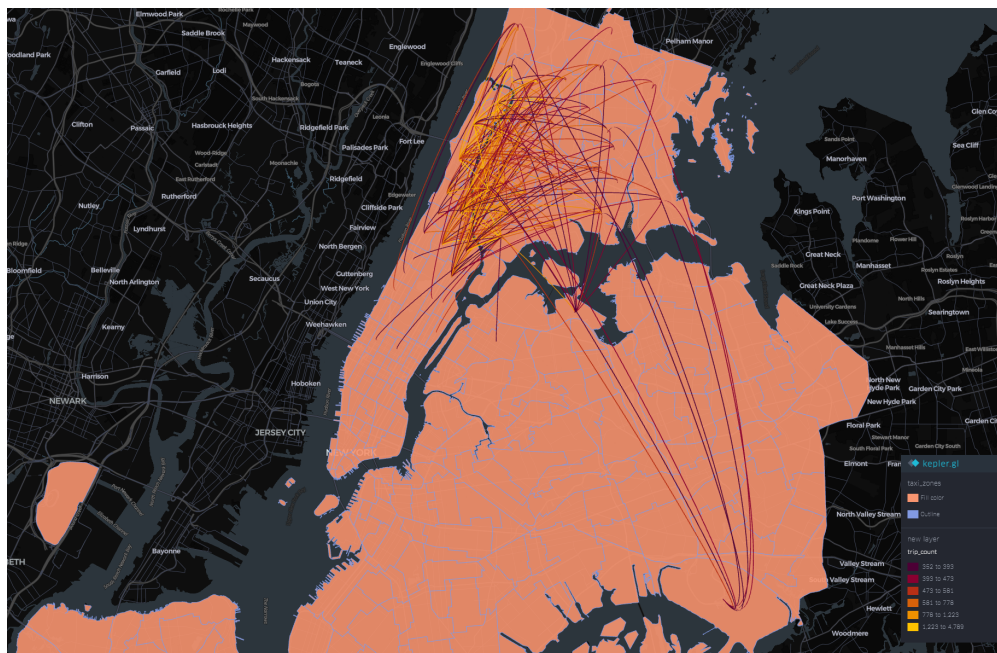


Figure 17: Trip counts out of the Bronx only including trips greater than 350

This trend underscores a dependency on taxi or FHV services for airport access due to limited public transit options that connect directly from Bronx neighborhoods to NYC airports.

5.3 2. Structural Bottlenecks: The Harlem River Problem

Now let’s look at the most trips into The Bronx

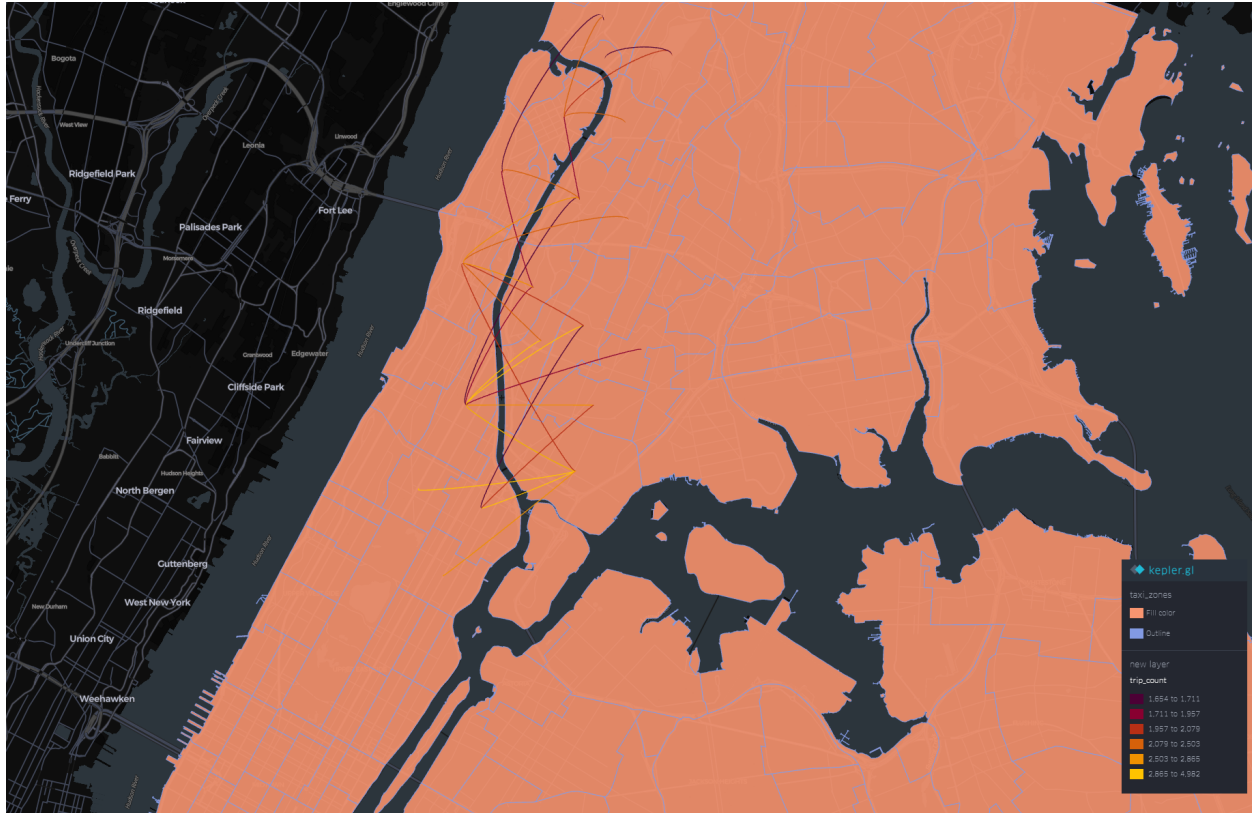


Figure 18: Trip counts to the bronx only including trips greater than 1632

The majority of non-local, non-airport trips from the Bronx travel **southwest out of Manhattan**, specifically across major bridges such as the University Heights Bridge and the Macombs Dam Bridge. A key insight from our spatial flow visualizations is that:

Trips are not just about distance they are about crossing points.

Our EDA suggests that **congestion and complexity in crossing the Harlem River are the most common driver of Bronx trip activity**. Passengers are effectively paying for reliable access across hard-to-navigate natural and infrastructural barriers. Further analysis will be needed to see how many trips are into Yankee Stadium, and the nearby hospital. But this section is extremely promising.

5.4 3. Vendor Usage Patterns

In the Bronx, usage of high-volume services such as Uber and Lyft dominates overall trip counts, especially for intra-borough and airport-bound trips. Traditional Yellow and Green taxis account for a smaller share of trips, often localized in higher-density commercial areas.

6 Conclusion

This exploratory data analysis combined, cleaned, and harmonized NYC taxi data into a unified geospatial framework to investigate mobility patterns, with a specific focus on the Bronx. After examining socioeconomic indicators and the spatial distribution of disadvantaged census tracts, we identified the Bronx as a key area of interest due to its high concentration of transit-vulnerable populations and limited transportation infrastructure.

The use of centroid mapping, borough filtering, and fare standardization provided a consistent basis for visualizing and quantifying travel behaviors within and beyond the Bronx. Our findings suggest that geographic constraints—rather than raw distance—are primary drivers of for-hire vehicle usage in the borough.

In particular, the Harlem River emerged as a major structural bottleneck, shaping trip volumes and destination patterns. Further analysis is needed to isolate the specific crossings and conditions that create this barrier. Future work will focus on quantifying these dynamics in greater detail and developing data-informed recommendations to improve mobility across this critical geographic divide.

References

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